Software Requirements Specification

AI Document Processing & Data Extraction System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the AI Document Processing & Data Extraction System, a B2B automation platform that converts unstructured business documents (invoices, receipts, purchase orders, resumes, contracts, and application forms) into structured, validated data. This document is intended to serve as the authoritative reference for the development team, project stakeholders, and quality assurance personnel throughout the design, implementation, and testing phases.

1.2 Document Conventions

Requirements use the prefix FR- (functional) or NFR- (non-functional), each with a priority of High, Medium, or Low. The keywords shall and must indicate mandatory requirements; should indicates a recommendation.

1.3 Intended Audience and Reading Suggestions

This document serves developers and architects (Sections 3, 4, 6, 7), project stakeholders (Sections 1–2), QA personnel (Section 3), and client reviewers (Sections 2 and 5).

1.4 Project Scope

The system automates the extraction of structured data from business documents that are otherwise processed manually. Employees currently spend significant time reading documents such as invoices and purchase orders and re-entering their contents into spreadsheets or accounting systems. This project replaces that manual effort with an automated pipeline: documents are uploaded, their type is classified, key fields are

extracted using an AI model, the extracted data is validated for consistency and duplication, and the results are made available through a searchable dashboard and exportable reports.

The long-term product vision, aligned with the project's six-stage roadmap, is a platform that can be customized per client with document-specific schemas and integrated into existing accounting and ERP systems. This SRS covers the full target feature set (OCR, validation, confidence scoring, human review, batch processing, dashboard, and export), while noting in Section 2.5 which capabilities belong to later iterations.

The system is not intended to replace human judgment on legally or financially significant decisions; all low-confidence or failed-validation extractions are routed to a human reviewer rather than being auto-approved.

1.5 References

Document	Description
Project Brief: AI Document Processing & Data Extraction System	Source project specification covering document types, architecture, roadmap, and monetization model
IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specifications, the structural template followed by this document

Test content.

2. Overall Description

2.1 Product Perspective

The AI Document Processing & Data Extraction System is a new, standalone, self-contained product. It is not a modification of an existing system, although it is designed to integrate with external accounting/ERP systems in later releases (Section 2.5). The system consists of four major subsystems: (1) a document ingestion and OCR layer, (2) an AI-driven classification and extraction engine, (3) a validation and duplicate-detection layer, and (4) a persistence, review, and reporting layer exposed through a web dashboard.

2.2 Product Functions

At a high level, the system shall:

- Accept single or batch uploads of PDF and image documents.
- Automatically determine each document's type (invoice, receipt, purchase order, resume, contract, or application form).
- Extract structured fields relevant to that document type using an AI model.
- Score the confidence of each extracted field and the document as a whole.
- Validate extracted data (e.g., arithmetic consistency of totals, presence of required fields).
- Detect duplicate documents, both by exact file match and by matching business key fields (e.g., same supplier and invoice number).
- Route low-confidence or failed-validation documents to a human reviewer for correction.
- Persist all documents and their extracted data in a relational database.
- Provide a searchable, filterable dashboard for reviewing documents and their status.
- Export structured data to Excel or CSV for downstream use.

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Data Entry / Operations Staff	Uploads documents daily and reviews flagged records	Low – uses the dashboard only
Reviewer / Approver	Corrects low-confidence extractions and approves records	Low to Medium
Administrator	Manages document-type schemas, user access, and system configuration	Medium
Developer / Integrator	Extends the system, adds new document schemas, builds ERP integrations	High
Business Owner / Client	Views reports and exported data; does not interact with the pipeline directly	Low

2.4 Operating Environment

The system shall operate as a containerized web application (Docker), deployable to a cloud provider or an on-premises VPS. The backend runs on Python 3.11+ with the FastAPI framework; the database is PostgreSQL 16 in production (SQLite is supported for local development only). The frontend dashboard is accessed through any modern evergreen web browser (Chrome, Edge, Firefox, Safari) and requires no client-side installation. OCR processing depends on the Tesseract OCR engine being available in the deployment environment.

2.5 Design and Implementation Constraints

- The system depends on a third-party or self-hosted Large Language Model (LLM) for the AI extraction step; extraction quality and latency are bounded by that model's capabilities and availability.
- Document-type schemas are defined in a central configuration module; adding a new document type does not require changes to the extraction, validation, or storage logic, but does require a code deployment in the current architecture (a future release may expose schema editing through the dashboard — see roadmap V6).
- OCR accuracy for scanned or low-quality documents is inherently limited by image resolution and document condition; the system mitigates this with confidence scoring and human review rather than guaranteeing perfect extraction.
- External ERP/accounting system integrations and per-client custom schemas are explicitly out of scope for the initial release (V1–V5) and are deferred to the roadmap's final stage (V6).

2.6 Assumptions and Dependencies

- It is assumed that uploaded documents are legitimate business documents in PDF or common image formats (PNG, JPG, TIFF) and not arbitrary file types.
- It is assumed that the organization deploying the system has, or will obtain, API access to a capable language model for the extraction step.
- The system assumes a single-tenant deployment per organization in the initial release; multi-tenant data isolation is a future consideration tied to the authentication work described in Section 5.2.
- It is assumed that reviewers possess sufficient domain knowledge to correct extracted fields (e.g., recognizing a misread invoice total) without additional in-app guidance beyond field labels and validation flags.

3. System Features / Functional Requirements

Requirements are grouped by pipeline module. Each requirement includes a priority: **H** (High – required for MVP), **M** (Medium), or **L** (Low – later roadmap stage).

3.1 Document Upload

ID	Requirement	Priority
FR-1	Upload a single document file (PDF, PNG, JPG, TIFF, or BMP) through the dashboard.	H
FR-2	Upload multiple documents in a single batch operation.	H
FR-4	Queue uploaded documents for background processing so the upload request returns immediately.	H

3.2 Text Extraction (OCR)

ID	Requirement	Priority
FR-6	Extract machine-readable text natively from digitally generated PDFs without invoking OCR.	H
FR-7	Detect pages with little or no extractable native text and automatically apply OCR to them.	H
FR-8	Apply OCR directly to image file uploads (PNG, JPG, TIFF, BMP).	H

3.3 Document Classification

ID	Requirement	Priority
FR-10	Classify each document into a supported type (invoice, receipt, purchase order, resume, contract, application form) or "generic."	H

ID	Requirement	Priority
FR-12	Fall back to AI-based classification when the heuristic method cannot classify with sufficient confidence.	M

3.4 AI Field Extraction

ID	Requirement	Priority
FR-14	Extract a defined set of fields per document type, per the central schema configuration.	H
FR-16	Assign a confidence score (0.0–1.0) to each extracted field.	H
FR-17	Compute an overall confidence score for the document.	H
FR-19	Gracefully handle extraction failures without crashing the pipeline.	H

3.5 Validation

ID	Requirement	Priority
FR-20	Check for missing required fields after extraction and record the result.	H
FR-21	Verify that subtotal plus tax equals total, within a small tolerance.	H
FR-23	Flag a document for human review when confidence is low or a check fails.	H

3.6 Duplicate Detection

ID	Requirement	Priority
FR-24	Detect exact duplicate file uploads via file hash comparison.	M
FR-25	Detect likely duplicate documents by comparing type-specific key fields (e.g., supplier + invoice number).	M

3.7 Human Review

ID	Requirement	Priority
FR-27	Let a reviewer view all extracted fields alongside validation results.	H
FR-28	Let a reviewer edit any extracted field value.	H
FR-29	Let a reviewer mark a document validated after review.	H

3.8 Batch Processing

ID	Requirement	Priority
FR-31	Process queued documents asynchronously in the background, independent of the originating request.	H

3.9 Dashboard and Search

ID	Requirement	Priority
FR-33	Display all documents with filename, type, status, confidence, and upload date.	H

ID	Requirement	Priority
FR-34	Filter the document list by status and document type.	H

3.10 Data Export

ID	Requirement	Priority
FR-37	Export filtered documents to Excel or CSV, including all extracted fields, status, confidence, and duplicate flag.	H

4. External Interface Requirements

4.1 User Interfaces

The system shall provide a browser-based dashboard with the following views:

- An upload view supporting drag-and-drop or file-picker selection, single or batch.
- A document list view with filter controls (status, document type), a search box, and summary statistics.
- A document detail view showing all extracted fields (editable), confidence scores, and validation check results.
- Export controls allowing the user to download the currently filtered document set as Excel or CSV.

The dashboard shall follow standard web accessibility practices (sufficient color contrast for status badges, keyboard-navigable form controls) and shall auto-refresh the document list periodically so that documents transitioning from "processing" to a terminal status are reflected without a manual page reload.

4.2 Hardware Interfaces

The system has no direct hardware interface requirements. It is deployed on standard server or cloud compute infrastructure; no specialized hardware (e.g., scanners, dedicated GPUs) is required, though a GPU may optionally accelerate OCR or a self-hosted extraction model if one is used.

4.3 Software Interfaces

Interface	Purpose
PostgreSQL (or SQLite for local development)	Persistent storage of document records, extracted fields, and validation results
Tesseract OCR engine	Text extraction from scanned documents and images
AI / LLM inference API	Structured field extraction and fallback document classification
REST API (FastAPI)	All dashboard-to-backend communication; also available for future ERP/accounting integrations

4.4 Communications Interfaces

All communication between the dashboard (client) and the backend occurs over HTTP(S) using JSON request and response bodies, except file uploads, which use multipart/form-data. Exported files are returned as binary file downloads (application/vnd.openxmlformats for Excel, text/csv for CSV). In production deployments, HTTPS with a valid TLS certificate shall be used to protect data, including any document content containing personal or financial information, in transit.

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-1	Respond to an upload request within 2 seconds, independent of background processing time.
NFR-2	Complete the end-to-end pipeline within 30 seconds for a typical single-page document.

5.2 Security

ID	Requirement
NFR-5	All client-server communication shall be encrypted using TLS in production.
NFR-6	Authentication and role-based access control shall be added before use with real business data (see Section 2.5).
NFR-7	Maintain an audit log of who uploaded, edited, or exported each document and when.
NFR-8	Uploaded documents and extracted data shall be encrypted at rest in production.

5.3 Reliability and Availability

ID	Requirement
NFR-10	A failure processing one document shall not affect other documents in the same batch.
NFR-12	The production deployment shall target 99.5% uptime, excluding maintenance windows.

5.4 Usability

ID	Requirement
NFR-13	A non-technical user shall be able to upload a document and locate its data without external training.

5.5 Maintainability

ID	Requirement
NFR-16	Document-type schemas shall be defined in a single, centralized configuration, not scattered across the codebase.

5.6 Scalability

ID	Requirement
NFR-18	The background processing architecture shall support horizontal scaling to increase throughput.

5.7 Compliance and Legal

ID	Requirement
NFR-20	The system shall be designed to support compliance with applicable data protection regulations (e.g., GDPR).
NFR-21	The system shall support permanently deleting a document and all derived data on request.

6. System Architecture Overview

The system is structured as a linear pipeline with a persistence and presentation layer at its end. A document moves through text extraction, classification, AI extraction, and validation before landing in storage, where it is either auto-validated or routed for human review.

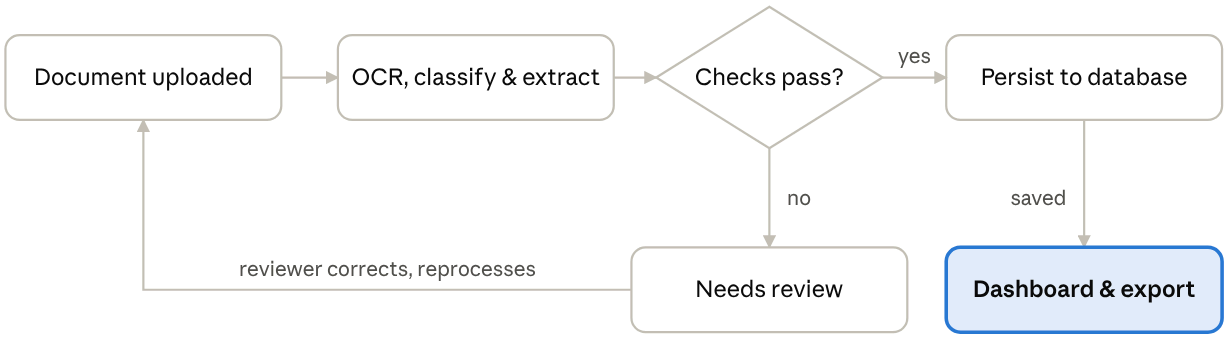
6.1 Technology Stack

Layer	Technology
Backend framework	Python 3.11+, FastAPI
Database	PostgreSQL (production), SQLite (local development)
PDF text extraction	PyMuPDF
OCR	Tesseract
AI extraction	LLM with structured JSON output
Data export	Pandas, openpyxl
Frontend	HTML/JavaScript dashboard (React/Next.js for a richer future UI)
Deployment	Docker, Docker Compose, cloud VPS

6.2 Pipeline Flow

The diagram below shows the path a single document takes from upload to its terminal status.

Validated documents reach the dashboard automatically



document pipeline · 5 steps, 1 decision

Documents flagged for review are corrected by a reviewer and reprocessed; validated documents proceed directly to the dashboard and are available for export.

7. Data Requirements

7.1 Core Entity: Document

The central entity in the system is the Document record, which holds both the raw upload metadata and the structured results of processing.

Field	Type	Description
id	UUID	Primary key
filename	String	Original uploaded filename
file_path	String	Storage location of the source file
file_hash	String	SHA-256 hash, used for exact-duplicate detection
doc_type	String	Classified document type (invoice, receipt, purchase_order, resume, contract, application_form, generic)
status	Enum	uploaded | processing | needs_review | validated | failed
raw_text	Text	Extracted text (native or OCR)
used_ocr	Boolean	Whether OCR was applied
extracted_fields	JSON	The structured field values, schema-dependent per doc_type
confidence_scores	JSON	Per-field confidence (0.0–1.0)
overall_confidence	Float	Aggregate confidence score
validation_results	JSON	List of validation checks and outcomes
is_duplicate_of	UUID (nullable)	Reference to the original document, if a duplicate
error_message	Text (nullable)	Populated when status is failed
created_at / updated_at	Timestamp	Record lifecycle timestamps

7.2 Document-Type Schemas

Each document type has its own field schema (e.g., an invoice's schema includes supplier_name, invoice_number, line_items, subtotal, tax, and total; a resume's schema includes candidate_name, skills, education, and work_experience). Schemas are centrally defined so that extraction, validation, and export logic all read from the same source of truth, and a new document type can be added without modifying pipeline code.

7.3 Data Retention

The system shall define a configurable retention period for source files and raw extracted text, after which they may be purged while retaining the structured field data needed for reporting (see NFR-9). Audit log entries (Section 5.2, NFR-7) are retained independently of the source document's retention period to preserve accountability history.

8. Appendices

8.1 Glossary

Term	Definition
OCR	Optical Character Recognition — converting an image of text into machine-readable text
Confidence score	A 0.0–1.0 value indicating how certain the AI extraction is about a field's correctness
Human review	Manual correction of extracted fields when confidence is low or a check fails
Schema	The defined set of fields the system extracts for a given document type

8.2 Use Case Summary

Use Case	Primary Actor	Goal
Upload a document	Operations staff	Submit a document for automated processing

Use Case	Primary Actor	Goal
Review a flagged document	Reviewer	Correct low-confidence fields and approve the record
Search and filter documents	Any dashboard user	Locate documents by status, type, or content
Export processed data	Business owner	Obtain structured data in Excel/CSV

8.3 Analysis Model

Section 6.2's process-flow diagram, together with the Document entity in Section 7.1, is the basis for the functional requirements in Section 3 and should guide database and API design.